WEIL RESTRICTION ALONG A FLAT PURE MORPHISM OF FINITE PRESENTATION IS REPRESENTABLE BY A CLOSED IMMERSION, OF FINITE PRESENTATION WHEN THE SUBSCHEME IS

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file flat.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the chapter's own retained minimal preamble (sources/stacks-flat/retained-preamble.tex, lines 1-85) is a verbatim excerpt of the native Xy-pic setup and of the native theorem declarations and mathematical macros, followed by the project's own reference/slogan/history display-environment definitions copied mechanically from the shared preamble; the wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler, and extra_packages is left empty because everything needed is already in the retained preamble. No macro was invented or redefined.

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Allowed standard prerequisites (editorial).

- The flattening map theorem, representing the locus where a map of modules becomes an isomorphism. Same chapter flat.tex, label theorem-flattening-map (invoked at proof line 6777). Applied to the surjection $u : \mathcal{O}_X \to \mathcal{O}_X/\mathcal{I}$, it represents by a closed immersion $Z \to S$ the T for which u_T is an isomorphism, which is exactly the Weil restriction functor F.
- The ideal sheaf of a closed subscheme and the associated surjection. Same chapter flat.tex, the ideal sheaf $\mathcal{I}$ of Y in X (used at proof lines 6772-6773). It converts the condition that $Y_T \to X_T$ be an isomorphism into the condition that the pullback u_T of the surjection be an isomorphism.

Source reference. Original theorem 05PF: sources/stacks-flat/pinned-original-source.tex, source lines 6671–6671. Navigation pointer for \ref{theorem-flattening-map} used inside the counted span (line 6777); the argument that consumes it is retained in the span.

Lemma 0.1. *Let $f : X \to S$ be a morphism of schemes which is of finite presentation, flat, and pure. Let Y be a closed subscheme of X. Let $F = f_*Y$ be the Weil restriction functor of Y along f, defined by*

$$F : (Sch/S)^{opp} \to Sets, \quad T \mapsto \begin{cases} \{*\} & \text{if} \quad Y_T \to X_T \text{ is an isomorphism,} \\ \emptyset & \text{else.} \end{cases}$$

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Then F is representable by a closed immersion $Z \to S$. Moreover $Z \to S$ is of finite presentation if $Y \to S$ is.

Proof. Let $\mathcal{I}$ be the ideal sheaf defining Y in X and let $u : \mathcal{O}_X \to \mathcal{O}_X/\mathcal{I}$ be the surjection. Then for an S-scheme T, the closed immersion $Y_T \to X_T$ is an isomorphism if and only if u_T is an isomorphism. Hence the result follows from Theorem 05PF. □